M. Minimum LCM

time limit per test 2 seconds
memory limit per test 512 megabytes
input standard input
output standard output

Rating: 1000 | Tags: math, number theory | https://codeforces.com/problemset/problem/1765/M

You are given an integer n.

Your task is to find two positive (greater than 0) integers a and b such that $a+b=n$ and the least common multiple (LCM) of a and b is the minimum among all possible values of a and b. If there are multiple answers, you can print any of them.

Input

The first line contains a single integer t ($1 \le t \le 100$) — the number of test cases.

The first line of each test case contains a single integer n ($2 \le n \le 10^9$).

Output

For each test case, print two positive integers a and b — the answer to the problem. If there are multiple answers, you can print any of them.

Example

Input
4 2 9 5 10

Output
1 1 3 6 1 4 5 5

Note

In the second example, there are 8 possible pairs of a and b:

- $a = 1$, $b = 8$, $LCM(1, 8) = 8$;
- $a = 2$, $b = 7$, $LCM(2, 7) = 14$;
- $a = 3$, $b = 6$, $LCM(3, 6) = 6$;
- $a = 4$, $b = 5$, $LCM(4, 5) = 20$;
- $a = 5$, $b = 4$, $LCM(5, 4) = 20$;
- $a = 6$, $b = 3$, $LCM(6, 3) = 6$;
- $a = 7$, $b = 2$, $LCM(7, 2) = 14$;
- $a = 8$, $b = 1$, $LCM(8, 1) = 8$.

In the third example, there are 5 possible pairs of a and b:

- $a = 1$, $b = 4$, $LCM(1, 4) = 4$;
- $a = 2$, $b = 3$, $LCM(2, 3) = 6$;
- $a = 3$, $b = 2$, $LCM(3, 2) = 6$;
- $a = 4$, $b = 1$, $LCM(4, 1) = 4$.